

HP at BAREC Shared Task 2026: Parameter-Efficient Ordinal Regression and Threshold Optimization for Arabic Readability.

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Abstract

This paper describes the system submitted by team HP for the BAREC Shared Task 2026 on Arabic readability assessment. Driven by the Quadratic Weighted Kappa (QWK) evaluation metric, which heavily penalizes distant misclassifications, we reframe the 19-level task as an ordinal regression problem rather than standard multi-class classification. Our approach leverages pre-trained language models (AraBERTv02, ARBERT) optimized via Parameter-Efficient Fine-Tuning (LoRA) and trained with a CONSistent RANk Log-its (CORAL) loss framework incorporating sample-level class weights. To ensure robust generalization, we consolidate the available datasets and implement a 5-fold stratified cross-validation pipeline. Finally, to mitigate validation overfitting and prediction shifting, we employ a 50/50 soft blending strategy. This averages the integer predictions derived from an 18-threshold Nelder-Mead optimizer and a Distribution Alignment method, yielding robust performance on the official blind test. Our solution is released in the Github¹.

1 Introduction

Assessing the readability of text is a crucial task in Natural Language Processing (NLP) that facilitates educational tools, text simplification systems, and accessible content generation. For Arabic, this is challenging due to its rich morphology, dialectal variations, and a scarcity of fine-grained datasets. The BAREC Shared Task 2026 addresses this gap by challenging participants to predict Arabic sentence readability across 19 finely graded levels (Elmadani et al., 2026a, 2025a,b; Habash et al., 2025).

Most traditional approaches model readability assessment as a standard multi-class classification problem, implicitly optimizing for exact match accuracy. However, this formulation is highly

suboptimal when the primary evaluation metric is Quadratic Weighted Kappa. Because QWK quadratically penalizes predictions that are far from the true label, a model must respect the relative ordering of the 19 classes. To directly align the learning signal with this metric, our system approaches the task as an ordinal regression problem. Through rigorous ablation studies comparing nominal classification and standard regression baselines, we empirically demonstrate the superiority of this ordinal framing.

Furthermore, while pre-trained language models (PLMs) have achieved state-of-the-art results, full fine-tuning is computationally expensive and prone to overfitting on imbalanced datasets. We address this by leveraging Parameter-Efficient Fine-Tuning, specifically Low-Rank Adaptation (Hu et al., 2021). To maximize labeled data utilization, we consolidate all provided splits into a 5-fold stratified cross-validation pipeline. Finally, to counteract conservative prediction behavior on unseen data, we develop a two-pronged post-processing strategy combining threshold optimization and distribution alignment, blended for optimal robustness.

2 Background

2.1 Task Setup and Dataset

The BAREC Shared Task 2026 focuses on fine-grained Arabic readability assessment at the sentence level (Elmadani et al., 2026a). The primary dataset provided is the BAREC Readability Corpus (Elmadani et al., 2025b; Altarouch et al., 2025; Elmadani et al., 2026b), which contains Arabic sentences annotated across 19 progressive readability levels based on strict annotation guidelines (Habash et al., 2025).

Our team participated in the **Strict Track**, which mandates that models be trained exclusively on the training set of the BAREC Corpus. Given the strict

¹<https://github.com/phonghoangKHDL/BAREC-HP-2026>

prohibition of external corpora, linguistic lexicons, or additional data augmentation, we focused our efforts on maximizing the potential of the provided data. We achieved this by fundamentally reframing the learning signal through ordinal regression rather than relying on external knowledge. Prior work in Arabic readability modeling has explored various feature engineering and neural architecture strategies (Liberato et al., 2024; Hazim et al., 2022).

2.2 Ordinal Regression and the QWK Metric

The primary evaluation metric for this shared task is the Quadratic Weighted Kappa, which measures the agreement between predicted and actual readability levels. Unlike standard accuracy, QWK is designed to heavily penalize distant misclassifications. Given $N = 19$ classes, the penalty weight $w_{i,j}$ between a predicted level i and an actual level j is defined quadratically as:

$$w_{i,j} = \frac{(i - j)^2}{(N - 1)^2} \quad (1)$$

Under this metric, confusing a Level 1 text with Level 5 incurs a penalty 16 times greater than confusing it with Level 2.

Standard multi-class classification objectives, such as Cross-Entropy or Focal Loss (Lin et al., 2018), assume all classes are mutually independent. They penalize a model identically regardless of whether its prediction is off by one level or eighteen levels, making them inherently mismatched with the QWK objective. On the other hand, standard regression losses (e.g., Mean Absolute Error) account for continuous distance but incorrectly assume that intervals between readability levels are perfectly equidistant, which rarely reflects true linguistic complexity.

Ordinal regression overcomes these limitations by acknowledging progressive difficulty without enforcing uniform spacing. By formulating the problem as a sequence of hierarchical binary boundaries via the CORAL framework (Cao et al., 2020), the model learns to respect rank order. This constraint naturally prevents extreme misclassifications, directly aligning the learning signal with the quadratic penalty mechanism of the QWK metric.

3 System Overview

Our proposed system architecture integrates deep text cleaning, Parameter-Efficient Fine-Tuning, and

an ordinal loss framework to address the specific challenges of the 19-level readability task.

3.1 Data Preprocessing and Contradiction Resolution

Real-world datasets often contain noise that can severely degrade the performance of fine-grained ordinal models. We applied a rigorous data cleaning pipeline. First, for **Text Normalization**, we utilized the ArabertPreprocessor to strip HTML tags, URLs, extra whitespaces, and Tatweel characters, ensuring standardized inputs (Elmadani et al., 2025b). Second, for **Label Contradiction Resolution**, we identified that identical sentences were occasionally assigned to different readability levels (hard contradictions). We discarded these conflicting instances to prevent optimization instability. Furthermore, exact text duplicates were removed to mitigate data leakage and accelerate training convergence.

3.2 Model Architecture and PEFT

We employed Arabic pre-trained language models, primarily AraBERTv02 and ARBERT, as our base encoders. Due to the relatively limited size of the training corpus and the vast parameter space of these models, full fine-tuning risks overfitting.

To counteract this, we integrated Low-Rank Adaptation (Hu et al., 2021). We injected trainable rank decomposition matrices into the query, value, and dense modules of the transformer layers. With hyperparameters set to rank $r = 16$, $\alpha = 32$, and a dropout of 0.05, we updated only a fraction ($\sim 1.7\%$) of the total parameters, significantly enhancing training stability and generalization.

3.3 Consistent Rank Logits Framework

To enforce ordinality, we replaced the standard classification head with the CORAL framework (Cao et al., 2020). Instead of predicting probabilities across 19 independent classes, the model outputs 18 binary logits. Each logit represents a binary classifier answering the question: “*Is the readability level strictly greater than k ?*” for $k \in \{0, 1, \dots, 17\}$.

Given a sentence x_i with a true label mapped to $y_i \in \{0, \dots, 18\}$, we generate a binary target vector T_i of length 18, where $T_{i,k} = 1$ if $y_i > k$, and 0 otherwise. The model outputs 18 logits $\hat{y}_i$. The loss for a single sample is computed as the

mean of 18 Binary Cross-Entropy (BCE) losses:

$$\mathcal{L}_{CORAL}^{(i)} = \frac{1}{18} \sum_{k=0}^{17} BCE(\hat{y}_{i,k}, T_{i,k}) \quad (2)$$

During inference, the final predicted level is obtained by applying a threshold (e.g., 0) to the 18 binary logits, summing the positive responses to obtain a value in $\{0, \dots, 18\}$, and then adding 1 to obtain the final predicted level in $\{1, \dots, 19\}$.

3.4 Sample-level Class Weights

The BAREC dataset exhibits class imbalance, which can bias the binary classifiers in the CORAL framework (Cao et al., 2020). We computed balanced class weights W based on the training distribution. To prevent gradient explosion, these weights were clipped to the range $[0.5, 5.0]$ and re-normalized. The final loss applies these weights at the sample level:

$$\mathcal{L}_{Final} = \frac{1}{N} \sum_{i=1}^N W_{y_i} \cdot \mathcal{L}_{CORAL}^{(i)} \quad (3)$$

4 Experimental Setup

4.1 Two-Phase Evaluation Strategy

To ensure robust generalization and prevent leaderboard overfitting, our experimental workflow was divided into two distinct phases. In the **Development Phase**, we first evaluated AraBERTv02 under various loss objectives using standard data splits, identifying the CORAL framework as optimal. We then implemented a 5-Fold Stratified Cross-Validation pipeline with multi-seed training to maximize data utilization, generating Out-Of-Fold predictions that significantly boosted our internal validation score. In the **Testing Phase** (for the blind test), we maintained this proven OOF methodology, training an additional ARBERT baseline for comparison, and focused our efforts on calibrating advanced post-processing techniques on the OOF logits without leaking test information.

4.2 Implementation Details

We utilized the Hugging Face transformers and peft libraries. Input sentences were tokenized with a maximum sequence length of 128. Models were trained for 5 epochs using a learning rate of 3×10^{-4} with a weight decay of 0.01. To accommodate memory constraints while maintaining a sufficient batch size, we used a per-device

batch size of 8 with 4 gradient accumulation steps. All ablation experiments shared this identical configuration to ensure a fair comparison. Training was conducted using the bfloat16 precision format, which provided improved numerical stability for the gradients, particularly for our primary CORAL loss framework.

5 Post-Processing and Ensembling

Our OOF experiments revealed that fine-tuned models trained on ordinal datasets often exhibit conservative prediction behaviors. To optimize the QWK metric for the final blind test, we implemented a three-step post-processing pipeline.

5.1 Nelder-Mead Threshold Optimization

Instead of relying on a default zero threshold for the 18 CORAL logits, we employed the Nelder-Mead optimization algorithm to empirically search for the 18 optimal decision boundaries. By directly maximizing the QWK score on our blended multi-seed OOF logits during the private phase, this method yielded an initial competitive test score of 0.8360. However, relying solely on rigid decision boundaries carries a risk of distributional overfitting on an unseen test set.

5.2 Distribution Alignment

To guard against prediction shifting and ensure statistical stability, we implemented a Distribution Alignment technique. First, we transformed the raw test logits into probabilities using the sigmoid function and summed them to compute a continuous difficulty score for each sentence. We then sorted the blind test sentences by this continuous score and assigned discrete integer labels (1 to 19) strictly following the class distribution (quotas) observed in the OOF training set. This effectively forces the model to predict minority classes that it might otherwise ignore.

5.3 Final Ensemble: 50/50 Soft Blending

To balance the direct metric optimization of the Nelder-Mead thresholds and the statistical safety of Distribution Alignment, we formulated a Soft Blending strategy. Our final submission averages the integer predictions produced by both methods (a 50/50 blend), reducing local inconsistencies and mitigating edge-case errors, yielding our most robust leaderboard performance.

Paradigm	Loss Function	QWK
Nominal Classification	Focal Loss ($\gamma = 2.0$)	0.7958
Nominal Classification	Weighted Cross-Entropy	0.8035
Ordinal Regression	EMD	0.8036
Standard Regression	Mean Absolute Error	0.8228
Ordinal Regression	CORAL Loss	0.8242

Table 1: Performance comparison of different loss objectives on the validation set.

6 Results and Discussion

6.1 Internal Validation and Ablation Studies

To evaluate our modeling choices and the contribution of different objective functions during the Development Phase, we conducted an ablation study on the validation set using AraBERTv02. Table 1 summarizes the internal QWK scores across various loss formulations prior to blind test evaluation.

The results highlight a crucial trade-off dictated by the QWK metric. Nominal classification models (Focal Loss (Lin et al., 2018) and Weighted Cross-Entropy) implicitly optimize for exact matches but fail to penalize distant predictions, resulting in lower QWK scores (0.7958 and 0.8035). While EMD (Hou et al., 2017) models cumulative distribution distances, it only achieved 0.8036.

Conversely, treating the problem as a continuous space via Mean Absolute Error provided a highly competitive QWK of 0.8228, as its distance penalty closely mirrors the QWK mechanism. However, MAE rigidly assumes equidistant intervals between levels. Our CORAL framework (Cao et al., 2020) overcomes this and remains superior (0.8242) by learning 18 flexible binary boundaries. Based on this result, we selected CORAL for our full 5-fold stratified cross-validation with multi-seed training, elevating our internal out-of-fold (OOF) validation score to **0.8523**.

6.2 Official Blind Test Results

In the Private Testing Phase, we applied our post-processing pipeline to the multi-seed OOF logits. We evaluated both base encoders under this identical setup. While the ARBERT backbone achieved a respectable official QWK of 0.8230 on the blind test, AraBERTv02 proved significantly stronger.

Specifically for AraBERTv02, applying the Nelder-Mead threshold optimization alone achieved a strong QWK of **0.8360**. To further prevent prediction shifting and improve stability, we introduced Distribution Alignment and formulated our **50/50 Soft Blend** strategy. By averaging

the integer predictions from both Nelder-Mead and Distribution Alignment, our AraBERTv02 submission gained an additional boost, reaching our top official score of **0.8370**. This outperformed other stacking variants (such as Ridge Regression and LightGBM), validating the generalization capability of our post-processing ensemble.

6.3 Error Analysis

An analysis of the out-of-fold confusion matrices revealed a prevalent issue: conservative prediction bias. Because QWK heavily penalizes distant errors, base models learned to safely cluster their predictions around the dataset’s majority classes (e.g., Levels 10, 12, and 14) to minimize expected loss, while severely under-predicting the extreme minority classes (e.g., Levels 18 and 19).

Our Distribution Alignment technique effectively mitigated this limitation. By enforcing exact label quotas based on continuous difficulty scores, we compelled the model to assign minority classes to sentences with the most extreme logits. This intervention substantially improved the recall for edge-case readability levels and ensured our final submission perfectly mirrored the real-world label distribution.

7 Conclusion

In this paper, we described our submitted system for the BAREC Shared Task 2026. By systematically evaluating multiple paradigms, we demonstrated that when the primary evaluation metric is QWK, reframing the Arabic readability assessment as an ordinal regression problem via the CORAL framework significantly outperforms accuracy-driven classification. Our pipeline showed that combining Parameter-Efficient Fine-Tuning with an ordinal hierarchical loss is highly effective. Furthermore, our Soft Blending strategy, which fuses Nelder-Mead optimization with Distribution Alignment, successfully countered the conservative prediction bias of ordinal models, achieving a robust QWK of 0.8370 on the blind test. Future work could explore integrating continuous regression anchors directly into the ordinal loss function to further refine extreme-level predictions.

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